

Lab: Probabilistic World Models for Robots

Objective

Design and analyze a probabilistic world model for a mobile robot performing simultaneous localization and mapping (SLAM). You will specify a generative model that captures the robot's beliefs about its environment, implement a factor graph representation, and compare world model architectures in terms of their free energy properties.

Prerequisites

- Understanding of Bayesian state estimation from Module 03
- Familiarity with occupancy grids or landmark-based maps
- Basic graph theory concepts (nodes, edges, factors)

Part 1: Generative World Model Design

Design a generative model for a robot exploring an unknown office environment:

1. **State variables:** Define the hidden states the robot must infer:
2. Robot pose trajectory: $\{x_0, x_1, \dots, x_T\}$
3. Landmark positions: $\{l_1, l_2, \dots, l_N\}$
4. Object categories: $\{c_1, c_2, \dots, c_N\}$ (e.g., door, chair, person)
5. **Generative process:** Specify the factored joint distribution:
6. $p(x_t | x_{t-1}, u_t)$: motion model conditioned on control input
7. $p(z_t | x_t, l_j)$: observation model for landmark j
8. $p(l_j)$: prior over landmark positions
9. $p(c_j)$: prior over object categories
10. How does this generative model enable the robot to simultaneously reason about "where am I?" and "what is around me?"

Write your response here

Part 2: Factor Graph Representation

Represent the SLAM generative model as a factor graph:

1. Draw or describe the factor graph with variable nodes (poses, landmarks) and factor nodes (motion model, observation model, priors).
2. Identify which message-passing algorithm could perform approximate inference on this graph (e.g., belief propagation, Gaussian elimination).
3. How does adding a loop closure observation (re-observing a known landmark) change the factor graph structure? What computational implications does this have?

Write your response here

Part 3: Hierarchical World Models

Robots operate at multiple spatial and temporal scales. Design a two-level hierarchical world model:

1. **Lower level:** Metric map -- precise positions and geometries (occupancy grid or landmark map).
2. **Upper level:** Topological map -- rooms, corridors, and connectivity (graph of places).

For each level: - What are the hidden states? - What is the timescale of updates? - How do predictions flow between levels (top-down priors, bottom-up evidence)?

Write your response here

Part 4: World Model Comparison

Compare three world model architectures for the same robot task (navigating from room A to room B):

Feature	Occupancy Grid	Landmark SLAM	Active Inference World Model
State representation			
Uncertainty handling			
Computational cost			
Adaptability			
Model complexity term			

Discuss how the free energy objective naturally balances model accuracy against model complexity (Occam's principle).

Write your response here

Part 5: Cognitive Map Update Pseudocode

Write pseudocode for updating the robot's world model upon entering a new room:

```
function update_world_model(belief, observation, world_model):  
    # 1. Predict expected observations from current world model  
    # 2. Compute prediction errors (what is surprising?)  
    # 3. Determine if existing model explains observations (low FE)  
    #    or if model structure must change (new room, new objects)  
    # 4. If model structure change needed:  
    #    - Add new landmark nodes to factor graph  
    #    - Initialize priors for new objects  
    # 5. Run inference to update all beliefs  
    return updated_world_model, updated_belief
```

Write your response here

Summary Table

World Model Component	Robotics Concept	Active Inference Concept	Example
Map	Occupancy grid / landmark set	Generative model structure	_____
Localization	Pose estimate	Posterior belief over robot state	_____
Object recognition	Classification	Model comparison / Bayesian model selection	_____
Novelty detection	Anomaly detection	High free energy (model surprise)	_____

References

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- Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1-2), 99-134.